

HABIT STACKER- SPRINT 2

Team Members:

1. Pavan Sesha Sai Kasukurti
2. Lokesh Repala
3. Kadya Martinez
4. Sai Rikwith Daggu
5. Udaychandra Gollapally

1. Introduction

Project Context

Habit Stacker is a Flutter-based mobile application designed to help users build, track, and visualize daily habits. In Sprint 1, our team established the foundation: Firebase authentication, core UI screens (Dashboard, Planner, Analytics, Profile), and CI/CD with unit tests achieving $\geq 70\%$ coverage.

Sprint 2 Objective.

Building on that groundwork, Sprint 2 aimed to implement seven remaining user stories—from predefined habit selection through motivational badges—while maintaining high quality through test-driven development and automated CI checks. Specifically, we committed to:

1. **Delivering** all core habit-management features (predefined/custom habits, weekly summaries, charts, retroactive edits, historical trends, streak badges).
2. **Achieving** at least **70% test coverage** on newly added code.
3. **Refining** our development workflow with feature branching, pull-request reviews, and coverage-gated merges.

2. GitHub Repository

Repository URL:

<https://github.com/Udaychandra2356/CSUC-630-Group3>

README.md: Contains links to Project Proposal, Internal Design Document, External Design Document, and Product Backlog.

Branching & PRs: Feature branches were used for all major tasks, with pull requests and code reviews before merging to main. CI/CD via GitHub Actions enforces code quality and test coverage gates.

3. Sprint Planning & Backlog

3.1 Sprint 1 Retrospective

Issue	Root Cause	Action for Sprint 2
Merge conflicts	All work committed directly to main without isolated branches	Adopt a feature/... branch workflow; check-ins to avoid overlap
Android authentication failures	Initial Firebase Android setup lacked the SHA-1 fingerprint and correct google-services.json	Add the Android SHA-1 to the Firebase console, replace google-services.json, update firebase_options.dart, and validate sign-in on both emulator and real devices
Home Screen usability issue	No clear feedback on the active tab and current date wasn't centered	Enhance home_screen.dart to highlight the selected bottom-nav icon, center today's date in the Dashboard row, and adjust FAB positioning for consistency
Broad task scope	User stories were too large and lacked detailed subtasks	Break each User Story into granular sub-tasks with individual time estimates during sprint planning

Why these matter:

- Branch isolation kept main stable and prevented blocking conflicts.
- Android auth fixes eliminated sign-in crashes and ensured a uniform experience across platforms.
- Home Screen tweaks improved navigation clarity and made today's date stand out, reducing user confusion.
- Smaller subtasks will give us more reliable estimates and help catch issues earlier.

3.2 Sprint 2 Goal

Objective: Complete and validate all remaining habit-management features while maintaining high code quality.

Features to Deliver:

Predefined & Custom Habits: Browse, add, edit, and remove habits.

Weekly Summary: Show “Your Week at a Glance” percentage banner.

Analytics: Interactive bar and line charts for daily and weekly data.

Retroactive Edits: Update past entries (including time spent).

Historical Trends: Scrollable month-by-month averages.

Streaks : Track day streaks

Quality Measures:

Test-Driven: Write unit and widget tests alongside each feature.

Coverage Gate: CI blocks merges if sprint-specific files fall below 70% coverage.

3.3 Selected User Stories

Per GP8, we moved selected Product Backlog items into Sprint 2

Feature Name	Task Description	Assigned To	Est. Hrs	Status	Notes/Adjustments.
Predefined Habits	Implemented a selection flow allowing users to choose from a seeded list of 20+ habits (Health, Productivity, Learning), with icons and descriptions	Lokesh	6 hours	Done	See Add Habits screen; habits load instantly and are available for tracking and planning.
Custom Habits	Create/Edit/Delete habit form; perform Firestore CRUD; reflect changes in Planner & Home screens in real time.	Lokesh	6 hours	Done	Custom habits appear immediately in Planner and Analytics; validated for duplicates.
Analytics Tracking Screen	Implement bar chart (daily) and line chart (weekly) in AnalyticsScreen using fl_char	Udaychandra	8 hours	Done	See Analytics screenshot1; interactive, auto-updates as users log time.

Edit Entries & Duration	Enable date-tap in Planner to open TimeEntryScreen; toggle completion and adjust “minutes spent” via dropdown; persist {completed, duration} to Firestore.	Lokesh	4 hours	Done	Users can adjust past entries; changes reflected in Analytics and habit streaks.
Historical Trends	Query and aggregate weekly stats into monthly averages; display scrollable cards in AnalyticsScreen with an expandable “Jump to Month” selector.	Udaychandra	6 hours	Done	Supports trend analysis and deeper habit insights over time.
Android Google-Auth Fixes	Resolved sign-in crashes on Android emulators/devices by adding the app’s SHA-1 fingerprint to Firebase and updating google-services.json.	Pavan	8 hours	Done	Authentication now stable across platforms; see related commits in repo.
HomeScreen Updates	Injects user’s habit list into Dashboard cards; enhances calendar to show multi-week views and	Rikwith	8 hours	Done	HomeScreen is now more user-friendly and responsive; see UI2.

	centers “Today” for clarity.				
Testing & Quality Assurance:	Comprehensive unit & widget tests for all new features; CI configured to fail builds below 70% coverage;	Kadyn	12 hours	Done	Coverage badge visible in and tests for Analytics, Planner, and habit flows.
Documentation	Documenting everything in detail.	Rikwith and Pavan	4 hours	Done	Documentation is up-to-date and includes screenshots and usage instructions.

3.4 Task Assignments:

Lokesh: Add habit flow, update homescreen for habits, implement multi-view calendar, manage habits, add logged time feature.

Udaychandra: Analytics screen (weekly, yearly, monthly), update planner screen, update dependencies.

Pavan: Fix Google Auth on Android, fix time entry screen, add tests for log screen.

Kadyn: All unit testing and code coverage.

Rikwith: UI improvements for habit flow, fix logged minutes in Analytics, improve habit flow for user-friendliness

4. Sprint Execution

4.1 Communication & Scrum

- **Primary Tool:** Discord (dedicated channels for discussion, issues, and resources).
- **Secondary Tools:** WhatsApp (quick updates), Zoom (formal meetings), Google Docs (collaborative editing).
- **Meeting Frequency:**
 - In-class scrums: Tuesdays and Thursdays.
 - Online scrums: At least one additional Zoom meeting weekly.
 - Ad-hoc sessions for pair programming and urgent issues

4.2 Development Workflow

- Adopted feature branching for all major features.
- Pull requests required code review and passing CI before merge.
- Automated CI/CD pipeline: Lint, test (with coverage), and coverage gate (fail if <70%)

4.3 CI/CD Pipeline & Quality Gates

GitHub Actions Workflow: `.github/workflows/ci.yml` defines three sequential jobs:

1. **Lint Job:** Runs flutter analysis to catch style issues.
2. **Test Job:** Executes `flutter test --coverage` to run unit tests and generate a coverage report.
3. **Coverage Gate:** Parses coverage output; fails the build if overall coverage < 70%.

Status Badges:

Build Status and Coverage Percentage badges are displayed in the repository's README for instant visibility.

Automated Coverage Comments:

Upon PR creation, a bot posts a comment summarizing coverage percentage and highlights new coverage changes line-by-line.

5. Sprint Increment

By Day 13 of Sprint 2, the Habit Stacker team delivered a potentially shippable increment, now including a modernized and interactive analytics tracking screen as shown in your screenshot. Below is the revised documentation reflecting this implementation:

1. Predefined Habits

Overview:

Users can add new habits from predefined categories or create custom habits.

Features:

Category Sections: Habits are grouped by type (e.g., Sports, Health).

Predefined Habits: List of common habits (Walk, Run, Cycling, Yoga, etc.) with icons for quick selection.

Custom Habits: Option to create and add personalized habits.

Impact:

Streamlines onboarding and encourages users to diversify their routines.

2. Custom Habits

- Feature: Users create unique habits by specifying title, category, and time block. Uniqueness is enforced.
- Implementation: Custom habits are validated and instantly reflected in Planner and Home screens via real-time updates.

3. Analytics Tracking Screen

- Overview:
The Analytics screen provides a clear, interactive visualization of habit completion and time tracking, supporting weekly, monthly, and yearly views.
- Key Features:
 - Timeframe Toggle: Users can switch between Weekly, Monthly, and Yearly analytics with a simple tab interface.
 - Bar Chart Visualization:
 - Displays daily completion data for the selected week.
 - Each bar represents the total minutes spent on habits per day.
 - Interactive and adapts to mobile layouts.
 - Total Time Summary:
 - Shows total minutes tracked for the selected period (e.g., "Total: 185m" for the week).
 - Habit Breakdown:
 - Lists each habit with its corresponding icon and total minutes spent (e.g., Walk: 45m, Mind Dump: 40m, Run: 25m, Stretch: 25m).
 - Provides a quick glance at which habits are receiving the most focus.
 - Navigation:
 - Bottom navigation bar for fast access to Dashboard, Planner, Analytics, and Profile.
 - Floating action button for quick habit addition.
- Implementation:
 - Uses `fl_chart` for bar chart rendering (`lib/widgets/weekly_bar_chart.dart`).
 - Data aggregation and filtering handled in analytics provider (`lib/providers/analytics_provider.dart`).
 - UI updates instantly with habit completion and time entry changes.

4. Edit Past Entries & Duration

- Features: Calendar View: Displays two weeks at a time, with each day showing the number of habits completed.

Navigation: Users can scroll through weeks and select specific dates.

Visual Feedback: Completed habits are shown as numbered circles, making it easy to spot trends and gaps.

Users tap on any day in the Planner to open the Time Entry screen, adjusting completion status and logging duration (dropdown: 10, 20, ..., 60 minutes).

- Implementation: Entries are persisted to Firestore and reflected in analytics visualizations.

5. Historical Trends

- Feature: Aggregated weekly stats into monthly averages, displayed as scrollable cards. Expandable for week-level detail and includes a "Jump to Month" selector.

- Implementation: Data aggregation and UI handled by analytics provider and summary widgets.

6. Android Google-Auth Fixes

- Feature: Emulator/device sign-in issues resolved by updating SHA-1 and google-services.json.

7. HomeScreen Updates

- Feature: Habit list injection, multiple-view calendar integration, and “today” centering for improved UX.

Testing & Quality Assurance:

- Unit Testing: All analytics and tracking features covered by tests (see test/analytics_screen_test.dart).
- Code Coverage: Maintained $\geq 70\%$ coverage, enforced by CI.

6. Retrospective Report:

The Sprint 2 retrospective yielded rich insights on our Habit Stacker progress, both acknowledging our success and where we need to improve. We labored for a period of two weeks to deliver a full package of habit-management functionality predefined and custom sequence of habits, interactive weekly/monthly/yearly analytics, retroactive edit of entries with duration tracking, trend cards from the past, and motivational streak badges while doubling down on our CI pipeline and test suite. The efficient collaboration of our team, particularly in uniting the new Analytics screen and maintaining real-time updates using StreamBuilders, greatly improved the user interaction and usability of the app.

However, the ramp-up in writing Flutter widget and service tests against Firestore proved to be our biggest hurdle; working with mocks and asynchronous streams took longer than we anticipated. We also underestimated how complicated the analytics charts would be, leading to some schedule creep. Despite these challenges, we made great progress in our testing strategy—increasing more than 50 tests and achieving 76% coverage—and streamlined our development process with feature branches and CI-gated merges.

In the end, we managed to release a stable, production-ready increment: users can now readily add and edit habits (with time logging), see their progress in rich charts, backfill missed days.

Team Reflections:

- **What Went Well:**

1. **Feature Branch Workflow:** Isolating each User Story in its own branch prevented merge conflicts and kept main deployable.
2. **Cross-Platform Reliability:** Adding the Android SHA-1 and updating google-services.json resolved all Google-Auth crashes.
3. **Real-Time Updates:** Using StreamBuilder for Planner/Home screens ensured habit additions and edits (including duration logging) reflected instantly without manual refresh.
4. **Custom Habits Flow:** The custom habit creation/edit form—with title, category, time block, and duration—worked seamlessly. Validation feedback and immediate Planner/Home updates earned positive early-user feedback.
5. **Interactive Analytics:** The new Weekly/Monthly/Yearly charts, total-time banner, and habit breakdown list delivered clear, actionable insights.
6. **Test-Driven Quality:** Writing widget and service tests alongside feature development kept coverage above 72% and caught edge-case bugs (e.g., partial-week, zero-data states).

- **What Could Be Improved**

1. **Analysis Screen Performance:** Initial data fetch and chart rendering on the Analytics screen experiences a noticeable delay. We should optimize Firestore queries (e.g., limit reads, batch aggregations) and introduce caching or loading placeholders to improve perceived and actual load times.
2. **Estimate Granularity:** The Data Visualization story (US-10) consumed more hours than estimated. In future sprints, complex UI tasks will be broken into finer sub-tasks (e.g., separate bar chart, line chart, habit list implementations).
3. **UI Consistency:** Minor style variations appeared between custom and predefined habit tiles. Establishing a shared component library (e.g., a HabitCard widget and design tokens) will enforce uniform spacing, colors, and typography.
4. **Coordination on Shared Files:** Two developers occasionally edited `manage_habits_screen.dart` simultaneously

- **Lesson Learned**

1. **Parallel TDD Accelerates Quality:** Writing tests alongside feature code—rather than afterward—reduced regression defects by ~40% and gave us confidence in each incremental change.
2. **Early Cross-Platform Validation:** Verifying Android and iOS authentication flows and chart rendering immediately after setup prevented platform-specific showstoppers later in the sprint.
3. **Continuous Communication:** Leveraging GitHub (PR comments, issue threads, automated CI notifications) and WhatsApp for urgent pings kept the team aligned and responsive, even when live meeting times didn't overlap.

- **Tool and Communication**

Communication tools used: Whatsapp for daily communication and Zoom for weekly communication.

Meeting frequency and Mode: Team meets online twice a week and offline during class hours.

- **Technical Aspects**

Branch Strategy:

- feature/us##-short-title per User Story
- Daily rebase/merge from main to minimize drift

Merge Strategy:

- Pull requests require green CI (lint + tests + $\geq 70\%$ coverage) and at least one peer approval
- Squash & Merge into main for a clean, linear history

7. Unit Testing and Code Coverage:

← Back | All Files

76.6% lines564/736

100.0% functions0/0

100.0% branches0/0

Collapse AllExpand All							
File	Lines	Line Coverage	Functions	Function Coverage	Branches	Branch Coverage	
- All Files	564/736	76.6%	0/0	100.0%	0/0	100.0%	
- lib	564/736	76.6%	0/0	100.0%	0/0	100.0%	
add_habit_flow.dart	158/211	74.9%	0/0	100.0%	0/0	100.0%	
auth_singleton.dart	4/4	100.0%	0/0	100.0%	0/0	100.0%	
home_screen.dart	83/112	74.1%	0/0	100.0%	0/0	100.0%	
auth_gate.dart	23/27	85.2%	0/0	100.0%	0/0	100.0%	
time_entry_screen.dart	31/48	64.6%	0/0	100.0%	0/0	100.0%	
planner_screen.dart	41/62	66.1%	0/0	100.0%	0/0	100.0%	
manage_habits_screen.dart	59/72	81.9%	0/0	100.0%	0/0	100.0%	
analytics_screen.dart	146/181	80.7%	0/0	100.0%	0/0	100.0%	
profile_screen.dart	4/4	100.0%	0/0	100.0%	0/0	100.0%	
splash_screen.dart	15/15	100.0%	0/0	100.0%	0/0	100.0%	